



iPod tricks being use to converse

Children are conversing in classrooms by swapping single iPods and using a text to speech app. <https://digit.in/feb20-37>



Hack GPS into a DSLR

Learn how to link up your phone's location data to your DSLR or mirrorless camera with Google Photos. <https://digit.in/feb20-38>

on VulnHub, making it a purely lab-focussed platform. Again, machines are available and approachable with users of any experience and machine difficulty is usually mentioned in the description. VulnHub is community-driven as all the vulnerable virtual machines are created by fellow users and approved by the moderating staff. Even though no officially maintained forum is present, VulnHub recently got their own discord server [<https://discord.gg/uHn4xdb>] where updates about new machines, write-ups, etc. are constantly posted. That's the first place we'll start looking for some help in case of getting stuck.

VulnHub provides a clutter free web interface to download machines to your liking. No sign-up or login is required, each machine comes with their defined

terface and easier configuration. Still, at the end of the day it's about what you feel comfortable working on.

A lot of other online lab platforms are available like PentesterLabs, VirtualHackingLabs, etc, but the most useful resources are either paid or are connected to some certifications they sponsor, hence, we chose the above two platforms which are best for beginners. Now let's take a look at the challenge based platforms.

PLATFORMS PROVIDING CHALLENGES:

Instead of simulating real-world applications, these platforms provide challenges to hone your skills in various sub-sectors of cybersecurity interests be it web application hacking, cryptog-

Basic challenges are mostly isolated and focussed on a specific set of skills. On the other hand, Realistic challenges may require knowledge from one or more domains. Some interesting categories like Phone-phreaking challenges, IRC (Internet Relay Chat) challenges are also present which are rarely seen on other platforms. Most of the challenges are created by the HTS staff and moderators while some are crafted by the members. Like the HackTheBox, a ranking system is used where you can attain the ranks in the following order,

Pentitioner->Script Kiddie->Apprentice->Wiseman->Master->Hacker->Elite->Sage

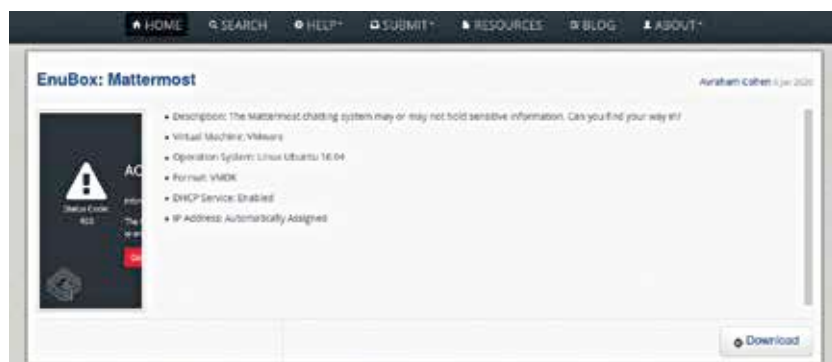
Ranks are determined by the number of points you have which you earn by completing the challenges. A special 'Hacker Emblem' is also there but it's only given to "those who deserve", according to the website.

A highly efficient forum is maintained where the challenges are discussed extensively. As the platform is quite old, you might not get your questions answered quickly but then again there's a 99% chance that what you are looking for has already been asked and solved, so make sure to check the forum thoroughly before posting. Other than posts about challenges, the forum is brimming with information ranging from hardware to gaming, book discussions to telecommunication. Being one of the older platforms, they haven't migrated to newer platforms like Discord or Slack but they are still active on their IRC channels, so remember to check out that if you feel the need. HackThisSite is strongly recommended if you are an absolute beginner as it's the right place to get the taste of the hacker culture.

2. Root-me

[<https://www.root-me.org>]

Like HackTheBox, Root-me is another hybrid platform providing both challenges and labs, but while HackTheBox is a lab-focussed platform providing challenges, Root-me is a challenge-focussed platform providing labs. The number of challenges present on Root-me is more than any other platform we



goals(either to uncover a certain amount of flags or gain root privileges). Here are some of our recommendations and general guidelines:

- VulnHub will be a better place to practice if you are on an unstable internet connection. You don't want to face delays due to other users doing the same machine you are doing. But then again, you can play around on both HTB and VulnHub.
- Keep your pentesting environment(the OS you'll be using to hack the machines) and the vulnerable machine on the same virtualizing software and same network settings (NAT, Bridged or Host-only).
- The best option is to keep both the systems as NAT, that has worked well for us in most cases.
- We recommend to use VMWare Player as it provides a more fluidic in-

terface, reverse engineering and or anything really! Challenges from different categories act as the building blocks for all numerous labs and practicing challenges is a great way to hone your skills.

1. HackThisSite!

[<https://www.hackthissite.org/>]

This has to be one of the oldest stomping grounds for hackers across the world. Created by Jeremy Hammond (hacker known by the handle "xec96", popular for whistleblowing and hacktivism) in 2003 as a "hacker training ground", it is a pretty respected website for acquiring the skills and mindset of a hacker.

Challenges are divided into 10 categories like Basic, Realistic, Forensics to name a few. We highly recommended beginners to start with the basic missions which will motivate you and prepare you to take on harder problems.